CRYPTOGRAPHY AND NETWORK SECURITY
Introduction to Symmetric Cryptography

Assignment 04

Q1. Explain the basic principles of symmetric cryptography. Describe plaintext, ciphertext, secret key, encryption and decryption algorithms, and Kerckhoffs's principle. Draw a labelled block diagram.

Meaning: Symmetric cryptography is a method of protecting information in which the sender and receiver use the **same secret key** for encryption and decryption. The key must be kept secret and shared safely between the two parties.

Main terms

- **Plaintext:** Original readable message or data. Example: “MEET AT 5”.
- **Ciphertext:** The unreadable form produced after encryption. It should not reveal the original message to an unauthorized person.
- **Secret key:** A value known only to the communicating parties. The encryption and decryption process depends on this key.
- **Encryption algorithm:** A mathematical procedure that converts plaintext into ciphertext using the secret key.
- **Decryption algorithm:** The reverse procedure that uses the same secret key to recover the original plaintext.

Working

1. The sender takes the plaintext and a secret key. 2. The encryption algorithm processes them and produces ciphertext. 3. The ciphertext is sent through the communication channel. 4. The receiver applies the decryption algorithm with the same secret key. 5. The original plaintext is recovered.

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Plaintext + Secret Key → Encryption Algorithm → Ciphertext → Decryption Algorithm + Same Secret Key → Plaintext
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Kerckhoffs's principle

Kerckhoffs's principle states that a cryptographic system should remain secure even if everyone knows how the system works. Only the key should need to remain secret. Security must not depend on hiding the algorithm. This makes systems easier to examine, test and improve.

Example: AES is a publicly known encryption standard. Its security depends on keeping the key secret, not on hiding the AES algorithm.

Conclusion: Symmetric cryptography is generally fast and useful for protecting large amounts of data, but secure key sharing and key management are important challenges.

Q2. Distinguish between stream ciphers and block ciphers. Explain confusion, diffusion and the modes ECB, CBC, CFB, OFB and CTR with strengths and weaknesses.

Stream cipher and block cipher

Basis	Stream cipher	Block cipher
Operation	Encrypts data one bit or byte at a time.	Encrypts fixed-size blocks of data.

Speed/use	Useful for continuous data streams.	Useful for files, disks and network data.
Padding	Normally does not need block padding.	May need padding, depending on the mode.
Example	RC4 (now considered insecure).	AES and DES.

Mode	Working	Strength
ECB	Encrypts each block independently with the same key.	Simple and parallel process
CBC	Each plaintext block is XORed with the previous ciphertext block before encryption. First block uses an IV.	Hides repeated block patter
CFB	Uses the previous ciphertext (and an IV initially) as input to the block cipher; output is combined with plaintext.	Can process data in smaller
OFB	Repeatedly encrypts an IV to create a keystream, which is XORed with plaintext.	Transmission errors do not s
CTR	Encrypts successive counter values to create a keystream; XORs it with plaintext.	Fast, parallelizable and supp

Confusion and diffusion

Confusion: Makes the relationship between the secret key and ciphertext difficult to understand. Substitution is commonly used to provide confusion.

Diffusion: Spreads the effect of one plaintext bit over many ciphertext bits, hiding patterns in the original message. Permutation and mixing operations provide diffusion.

Block cipher modes of operation

A mode explains how a block cipher is applied to messages longer than one block.

Important: ECB, CBC, CFB, OFB and CTR provide confidentiality modes. They do not automatically provide message authentication or integrity; authenticated encryption or a separate authentication mechanism is needed.

Q3. Trace the historical development of symmetric encryption algorithms.

Symmetric encryption developed from simple manual techniques into strong computer-based algorithms.

1. Early classical ciphers

The **Caesar cipher** was used in ancient Rome. Each letter is shifted by a fixed number of positions. It is easy to use but can be broken by trying the small number of possible shifts.

2. Enigma machine

During World War II, Germany used the Enigma machine. It used rotating wheels (rotors) and electrical connections to create changing letter substitutions. Allied codebreakers, including teams at Bletchley Park, helped break Enigma messages. This showed the importance of mathematics, machines and cryptanalysis.

3. Shannon's contribution

Claude Shannon's work on information theory and secrecy in the 1940s provided a mathematical foundation for modern cryptography. He described important design ideas such as **confusion** and **diffusion**.

4. Feistel structure

The Feistel structure divides a data block into two halves and processes them through repeated rounds. A round function and round keys are used. The same basic structure can be used for encryption and decryption by reversing the order of round keys.

5. DES in 1977

The Data Encryption Standard (DES) was adopted as a U.S. federal standard in 1977. It uses 64-bit blocks and a 56-bit effective key. DES became widely used, but its key later became too short to resist modern brute-force attacks.

6. Triple DES and other algorithms

Triple DES (3DES) applies DES three times to increase security. Other symmetric algorithms included **IDEA** and **Blowfish**. **RC4** was a stream cipher used in some older systems, but it has serious security weaknesses and should not be used.

7. AES competition

As DES became less secure, NIST organized an open competition to select a replacement. The Rijndael algorithm, designed by Joan Daemen and Vincent Rijmen, was selected and became the **Advanced Encryption Standard (AES)** in 2001. AES supports 128-bit blocks and 128-, 192- and 256-bit keys.

Conclusion: The development moved from simple letter substitution to mathematically designed algorithms such as DES and AES, which are suitable for modern digital communication.

Q4. Explain Caesar, monoalphabetic substitution, Playfair and Vigenere ciphers. Why is frequency analysis effective against some but not others?

(a) Caesar cipher

Each plaintext letter is shifted by a fixed number of places in the alphabet. For a shift of 3: A becomes D, B becomes E and so on. Example: HELLO becomes KHOOR. It is simple but weak because there are only 25 useful shifts.

(b) Monoalphabetic substitution cipher

Each letter is replaced by another fixed letter according to one substitution alphabet. For example, A may always become Q and B may always become W. It has more possible keys than Caesar, but the same plaintext letter always maps to the same ciphertext letter. This preserves letter-frequency patterns.

(c) Playfair cipher

Playfair encrypts pairs of letters (digraphs) using a 5 × 5 key matrix, usually combining I and J. The message is split into pairs. If both letters are in the same row, take the letters to their right; if in the same column, take the letters below; otherwise, form a rectangle and take the letters in the same rows at the opposite corners. Repeated letters in a pair are separated and a filler such as X may be added.

Strength: It hides single-letter frequency better than simple substitution. **Weakness:** It can still be attacked using digraph (letter-pair) frequency and other statistical methods.

(d) Vigenere polyalphabetic cipher

Vigenere uses a repeating keyword to apply different Caesar shifts to different letters. The same plaintext letter can become different ciphertext letters depending on the key letter. Example: a keyword controls the shift used for each position.

Strength: It hides ordinary single-letter frequency better than a single fixed substitution. **Weakness:** If the keyword repeats, key length and statistical patterns may be discovered.

Frequency analysis

Frequency analysis studies how often letters or groups of letters occur. In English, E, T and A are common letters. Caesar and monoalphabetic substitution preserve the frequency pattern under renamed letters, so they are vulnerable. Playfair works on pairs, making simple single-letter analysis less useful. Vigenere uses multiple substitutions, so the pattern is spread across different alphabets; however, repeated-key patterns can still be analysed.

Q5. Describe rail fence and columnar transposition, product ciphers, and the one-time pad.

Transposition cipher

A transposition cipher changes the positions of letters without changing the letters themselves. The same letters remain in the ciphertext, but their order is rearranged.

(a) Rail fence cipher

The message is written diagonally across a set number of rows (rails) and then read row by row. Example with two rails, the letters of a message alternate between the top and bottom rows. It is easy to understand but can be broken by trying different rail counts.

(b) Columnar transposition cipher

The plaintext is written in rows under a keyword. The columns are then read in the order determined by the alphabetical order of the keyword letters. It rearranges positions but does not replace letters. It is stronger than a very simple rail fence but can still be attacked using pattern and language analysis.

Product cipher

A product cipher combines substitution and transposition. Substitution changes the symbols, while transposition changes their positions. Repeating these operations creates confusion and diffusion. Modern block ciphers use multiple rounds of substitution, permutation and mixing, based on this general idea.

One-time pad (OTP)

A one-time pad combines the plaintext with a truly random key of the same length, commonly using XOR for binary data. The key must be used only once and kept completely secret.

- **Why it is theoretically unbreakable:** If the key is truly random, as long as the message, secret and used only once, every possible plaintext is equally consistent with the ciphertext. The ciphertext alone gives no information about the plaintext.

- **Why it is impractical:** A key as long as the message must be generated, securely delivered, stored and protected. Reusing a pad destroys its security. Managing large pads is difficult.

Q6. Describe the structure of DES, including Feistel structure, block/key size, permutations, 16 rounds, round function and key schedule.

DES (Data Encryption Standard) is a symmetric block cipher based on a Feistel network.

Main features

- Block size: 64 bits.

- Key supplied: 64 bits, of which 8 bits are used for parity; effective key length is 56 bits.

- Number of rounds: 16.

- Structure: Feistel network, dividing the block into left and right 32-bit halves.

DES working steps

1. **Initial permutation (IP):** The 64-bit input block is rearranged according to a fixed table. 2. **Split:** The result is divided into L0 and R0, each 32 bits. 3. **Sixteen rounds:** Each round uses a different 48-bit round key. 4. **Swap and combine:** After the rounds, the halves are arranged as specified by DES. 5. **Final permutation (IP■¹):** The inverse initial permutation is applied to produce the 64-bit ciphertext.

One DES round

For round i: Li = Ri; Ri = Li XOR F(Ri, Ki), where Ki is the 48-bit round key and F is the round function.

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Right half (32 bits) → Expansion (48 bits) → XOR with round key (48 bits) → S-boxes (32 bits)
    → P-box permutation (32 bits) → XOR with left half. The old right half becomes the new
                                        left half.
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Round function F

- **Expansion:** Expands the 32-bit right half to 48 bits.

- **Key mixing:** XORs the expanded data with the 48-bit round key.

• **S-boxes:** Eight substitution boxes convert 48 bits into 32 bits. They provide non-linearity and confusion.

• **P-box permutation:** Rearranges the 32 output bits to spread their influence and provide diffusion.

Key schedule

The 56 effective key bits are rearranged and split into two 28-bit halves. The halves are circularly shifted by one or two positions in each round. A selection/permutation step then chooses 48 bits to form each round key. This produces 16 round keys.

Decryption: DES uses the same Feistel structure, but applies the round keys in reverse order, from K16 to K1.

Q7. Discuss DES strengths and weaknesses, brute-force attacks, differential and linear cryptanalysis, Double DES and Triple DES.

Strengths of DES

• DES has a clear and well-studied 16-round Feistel structure.

• Its S-boxes and permutations provide confusion and diffusion.

• It shows the avalanche effect: a small change in plaintext or key can change many ciphertext bits.

Weaknesses of DES

Short key: DES has only 56 effective key bits. Modern computers can search this key space, so brute-force attacks are practical with sufficient computing resources.

Brute-force attack: An attacker tries possible keys until meaningful plaintext is found. The small key length is the main reason DES is no longer secure.

Differential cryptanalysis: Studies how differences in selected plaintext pairs affect differences in ciphertext. It can reveal information about the key.

Linear cryptanalysis: Uses statistical relationships between plaintext, ciphertext and key bits to gain information about the secret key.

Variant	Keying option	Description
Two-key 3DES	K1 and K2; the third operation reuses K1.	Provides a larger effective key space than DES, but less than three independent keys
Three-key 3DES	K1, K2 and K3 are independent.	Uses three independent keys and provides a stronger security margin than DES.

Double DES

Double DES applies DES encryption twice, usually with two keys: C = E(K2, E(K1, P)). It appears to provide a longer key, but it is vulnerable to a **meet-in-the-middle attack**. The attacker encrypts the plaintext with possible first keys and decrypts the ciphertext with possible second keys, then looks for a matching intermediate value. This reduces the expected security gain.

Triple DES (3DES)

Triple DES applies DES three times, commonly in Encrypt–Decrypt–Encrypt (EDE) form. It was designed to improve security while retaining compatibility with DES.

Conclusion: 3DES improved on DES but is slower than modern algorithms and is now considered legacy. AES is preferred for new systems.

Q8. Explain AES: block size, key lengths, rounds, four transformations, key expansion, decryption and comparison with DES.

Advanced Encryption Standard (AES) is a widely used symmetric block cipher selected by NIST. It is based on the Rijndael algorithm.

Main features

• Block size: 128 bits (fixed).

• Key lengths: 128, 192 and 256 bits.

• Number of rounds: 10 rounds for a 128-bit key, 12 for a 192-bit key and 14 for a 256-bit key.

Four round transformations

• **1. SubBytes:** Each byte is replaced using a fixed substitution table called the S-box. It provides non-linearity and confusion.

• **2. ShiftRows:** Rows of the state are shifted cyclically by different offsets. It helps spread byte positions.

• **3. MixColumns:** Bytes in each column are mathematically mixed. It provides diffusion. This step is omitted in the final encryption round.

• **4. AddRoundKey:** The state is combined with a round key using XOR.

AES encryption process

The 128-bit input is arranged as a 4 × 4 byte state. First, an initial AddRoundKey is performed. Then the main rounds apply SubBytes, ShiftRows, MixColumns and AddRoundKey. In the final round, MixColumns is omitted. The result is the ciphertext.

Basis	DES	AES
Block size	64 bits	128 bits
Key size	56 effective bits	128, 192 or 256 bits
Structure	16-round Feistel network	Substitution-permutation network
Rounds	16	10, 12 or 14
Security	Not secure against modern brute-force attacks due to short key.	Strong modern standard when correctly implemented and managed
Performance	Older design; generally less suitable for modern use.	Efficient in software and hardware; widely used.

Key expansion

AES uses a key expansion process to generate a separate round key for each round from the original secret key. The number of round keys depends on the selected key length. This process uses operations such as byte substitution, rotation, round constants and XOR.

AES decryption

Decryption reverses the encryption transformations using inverse operations: InvShiftRows, InvSubBytes, AddRoundKey and InvMixColumns. The round keys are applied in reverse order. The final inverse round omits InvMixColumns.

AES vs DES

Conclusion: AES is more secure and generally more suitable for modern applications than DES because it has a larger block size, longer keys and an efficient design.

QUICK REVISION – IMPORTANT FACTS

Topic	Remember
Symmetric cryptography	Same secret key is used for encryption and decryption.
Kerckhoffs's principle	Algorithm may be public; key must remain secret.
Confusion	Hides the relationship between key and ciphertext.

Diffusion	Spreads the influence of plaintext bits.
ECB	Same plaintext block → same ciphertext block; patterns may show.
CBC	Uses previous ciphertext block and an IV.
CFB / OFB / CTR	Modes that generate/extend stream-like encryption from a block cipher.
Caesar cipher	Fixed alphabet shift.
Playfair	Encrypts pairs of letters using a 5 × 5 matrix.
Vigenere	Uses repeating keyword for changing shifts.
One-time pad	Truly random key, same length as message, used once.
DES	64-bit block, 56-bit effective key, 16 rounds.
DES round	Expansion → XOR key → S-box → P-box.
AES	128-bit block; 128/192/256-bit keys; 10/12/14 rounds.

Exam tip: For long answers, write the definition first, then features or steps, add a labelled diagram/table where asked, and finish with strengths, weaknesses or a short conclusion.